

Controlling the Heat Dissipation in Temperature-Matched Plasmonic Nanostructures

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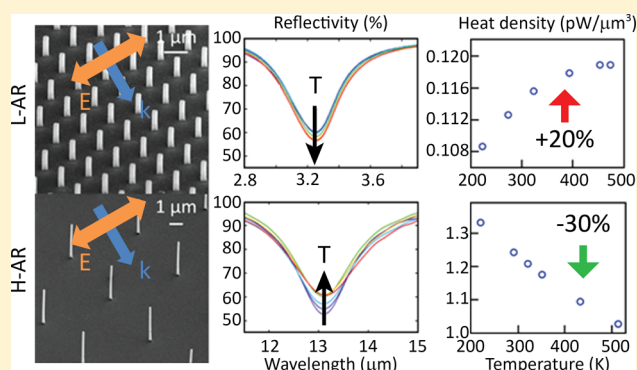
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Supporting Information

ABSTRACT: Heat dissipation in a plasmonic nanostructure is generally assumed to be ruled only by its own optical response even though also the temperature should be considered for determining the actual energy-to-heat conversion. Indeed, temperature influences the optical response of the nanostructure by affecting its absorption efficiency. Here, we show both theoretically and experimentally how, by properly nanopatterning a metallic surface, it is possible to increase or decrease the light-to-heat conversion rate depending on the temperature of the system. In particular, by borrowing the concept of matching condition from the classical antenna theory, we first analytically demonstrate how the temperature sets a maximum value for the absorption efficiency and how this quantity can be tuned, thus leading to a temperature-controlled optical heat dissipation. In fact, we show how the nonlinear dependence of the absorption on the electron–phonon damping can be maximized at a specific temperature, depending on the system geometry. In this regard, experimental results supported by numerical calculations are presented, showing how geometrically different nanostructures can lead to opposite dependence of the heat dissipation on the temperature, hence suggesting the fascinating possibility of employing plasmonic nanostructures to tailor the light-to-heat conversion rate of the system.

KEYWORDS: Nanoplasmonics, nanoscale heating, nonlinear effects, opto-thermal properties



Metallic nanostructures interacting with electromagnetic (EM) fields have been proven to be efficient light-to-heat converters^{1–18} both in continuous and pulsed regime.¹⁹ Heat dissipation by metallic nanostructures has been thus widely investigated and has found applications in different fields such as photonics,^{20–30} chemistry,^{31–36} medicine,^{37–42} and optofluidics.^{43–45} Furthermore, because different applications may require distinct temperature regimes, the interaction between metallic structures and EM fields has been studied both at low-^{46,47} and high-temperature operative conditions.^{48–50} In this context, the main parameter that is directly affected by temperature is the absorption damping coefficient Γ_a , which describes how the electron energy is dissipated within a metallic system upon electromagnetic excitation.⁵¹ One of the characteristics of the absorption damping coefficient is its approximately linear increasing value with growing temperature,^{50,52} an effect mainly due to electron–phonon scattering mechanism as described by Holstein,^{53,54} McKay, and Rayne.⁵¹

In this regard, by considering the temperature dependence of the damping mechanism in bulky metals, it has been explained the occurring decreasing reflectivity upon temperature increase within metallic surfaces.⁵⁵ In the following, we theoretically and experimentally prove that switching from flat to patterned plasmonic surfaces makes it possible to freely tune the light-to-heat conversion rate, allowing absorption and reflection to either increase or decrease by exploiting the rise of temperature. This concept can find direct applications in optoelectronics where thermal management is frequently an important aspect of modern low-dimension devices.

The role played by the absorption damping coefficient in determining the heat dissipation (i.e., absorption and scattering efficiencies) can be clarified by modeling the metal–photon

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interaction through an N charges harmonic oscillator.⁵⁶ The simplest case involves N electrons oscillating in a one-dimensional (1D) system (e.g., x -direction) due to the driving force associated with the electric field component E of a plane EM wave. This hypothesis is especially suitable for mimicking elongated plasmonic nanoantennas, a valid description as long as the charges oscillate at frequencies far from the intrinsic absorption of the material forming the antenna, that is, away from interband transitions. The resulting harmonic equation is

$$Nm_e\ddot{x} + N\Gamma_a m_e \dot{x} + Nm_e \omega_0^2 x - N^2 \Gamma_s \ddot{x} = NeE \quad (1)$$

Here m_e is the electron mass, Γ_a represents the absorption damping coefficient that formally plays the role of viscous friction, and ω_0 is the natural frequency of the system due to restoring forces, which for the present case of plasmonic nanostructures corresponds to the plasmon frequency (LSPR).⁵⁷ On the other hand, the term $\Gamma_s \ddot{x}$ represents the Abraham-Lorentz force, which expresses the recoil force a charge is subject to due to the emission of radiation while changing its acceleration,⁵⁸ with $\Gamma_s = \frac{e^2}{6\pi\epsilon_0 c^3}$ being the radiation damping coefficient. Finally, the term on the right-hand side of eq 1 corresponds to the driving force generated by a monochromatic plane wave, whose electric field $E = E_0 e^{i\omega t}$ is polarized along the x -axis. The general solution of the harmonic equation can therefore be written as $x = x_0 e^{i\omega t}$, where x_0 is the oscillation amplitude. At resonance, that is, $\omega = \omega_0$, the squared modulus of this quantity reduces to

$$|x_0|^2 = \left[\frac{\frac{eE_0}{m_e}}{\omega_0 \left(\Gamma_a + \omega_0^2 \frac{N}{m_e} \Gamma_s \right)} \right]^2 \quad (2)$$

Using this result, we can calculate both the average absorbed (P_a) and scattered (P_s) powers by the set of moving charges as⁵⁶

$$P_a = \frac{1}{2} N m_e \Gamma_a \omega_0^2 |x_0|^2 = \frac{1}{2} \Gamma_a \frac{\frac{Ne^2}{m_e}}{\left(\Gamma_a + \omega_0^2 \frac{N}{m_e} \Gamma_s \right)^2} E_0^2$$

$$P_s = \frac{1}{2} N^2 \Gamma_s \omega_0^4 |x_0|^2 = \frac{1}{2} \Gamma_s \frac{\left(\frac{Ne}{m_e} \right)^2}{\left(\Gamma_a + \omega_0^2 \frac{N}{m_e} \Gamma_s \right)^2} \omega_0^2 E_0^2 \quad (3)$$

From eq 3, we can readily make few observations. First, P_a and P_s depend on both the damping coefficients Γ_a and Γ_s . Second, whereas P_s monotonously increases as the absorption damping coefficient decreases, that is, it is maximized by $\Gamma_a = 0$, P_a has a more complex dependence on Γ_a . Furthermore, while Γ_s depends exclusively on universal constants, Γ_a is connected to tunable environment conditions such as temperature.⁵² Thus, its role becomes fundamental to control the light-to-heat conversion.

For simplicity, we now switch to the intensive parameters η_a and η_s , corresponding to the absorption and scattering efficiencies, respectively. Being $I_0 = \frac{c}{2} \epsilon_0 E_0^2$ the incident wave intensity and σ_g the geometrical cross section of the structure, these quantities are defined as $\eta_a = \frac{P_a}{\sigma_g I_0}$ and $\eta_s = \frac{P_s}{\sigma_g I_0}$. In the following, where cylindrical antennas are considered, the

geometrical cross section is $\sigma_g = 2RH$ with R and H being the base radius and the height of the cylinder, respectively. Interestingly, it is possible to find a value of Γ_a (here defined as Γ_a^M) for which the absorption efficiency reaches a maximum (η_a^M). In fact, by imposing $\frac{\partial \eta_a}{\partial \Gamma_a} = 0$, we have

$$\Gamma_a^M = \frac{N \Gamma_s \omega_0^2}{m_e} \quad (4)$$

We observe that Γ_a^M depends on the resonant frequency and the number of involved oscillating charges N (see the Supporting Information for the derivation of the expressions of both the absorption and scattering coefficients). By substituting eq 4 into eq 3, we can obtain η_a and η_s evaluated at Γ_a^M

$$\eta_a^M \equiv \eta_a|_{\Gamma_a^M} = \frac{1}{\sigma_g} \frac{3}{8\pi} \lambda_0^2 = \eta_s|_{\Gamma_a^M} \quad (5)$$

where $\lambda_0 = \frac{2\pi c}{\omega_0}$ is the resonant wavelength. From this equation, some important information can be deduced: (i) the maximum absorption that a plasmonic structure can achieve depends only on the resonant wavelength and on the geometric cross section of the system; (ii) at Γ_a^M , the absorption efficiency is equal to the scattering efficiency. On the other hand, the maximum scattering efficiency η_s^M is instead reached at $\Gamma_a = 0$, as shown in eq 3, and its value is

$$\eta_s^M = \eta_s|_{\Gamma_a=0} = 4\eta_a^M = \frac{1}{\sigma_g} \frac{3}{2\pi} \lambda_0^2 \quad (6)$$

It is important to note that this result had been found also in other contexts, both in classical and quantum electrodynamics. In fact, $\frac{3}{2\pi} \lambda_0^2$ corresponds to the total scattering cross section of an electron classically bound to an origin while interacting with a monochromatic wave oscillating near its restoring force natural frequency.⁵⁹ Furthermore, it describes also the cross section of an atom scattering a photon at the frequency corresponding to the energy difference between its ground and first excited state.⁶⁰ This is not surprising because we have modeled elongated antennas (e.g., plasmonic nanorods) as a harmonic oscillator where the restoring force natural frequency is the classical counterpart of the energy difference between two states within a quantum mechanical approach.

Similar considerations apply to eq 5 that can be contextualized in the classical antenna theory as well^{61,62} where it is used to describe the case of ideal macroscale lossless antennas. In particular, it describes a situation where the terminal resistance R_a equals the radiation resistance R_s ⁶³ resulting in the equivalence between absorbed and reradiated or scattered power, namely $P_a = P_s$. In our case this situation is expressed by $(\eta_a = \eta_s)|_{\Gamma_a^M}$. Indeed, a receiving antenna in this condition is said to be matched and it is employed to maximize the power absorbed by the load.⁶¹ On the other hand, the condition reported in eq 6 recalls to the short circuit case where reflection is maximized and no power is absorbed by the load: $R_a = 0$ and $P_a = 0$. In the present case it corresponds to $\Gamma_a = 0$ therefore $\eta_a = 0$ and $\eta_s = 4\eta_a^M$.

In this regard, we shall show that plasmonic nanorods (nanoscale plasmonic antennas) play a similar role of macroscale antennas, as schematized in Figure 1 and explained in details in Supporting Information. Indeed, by modifying the intrinsic absorption damping coefficient Γ_a through its

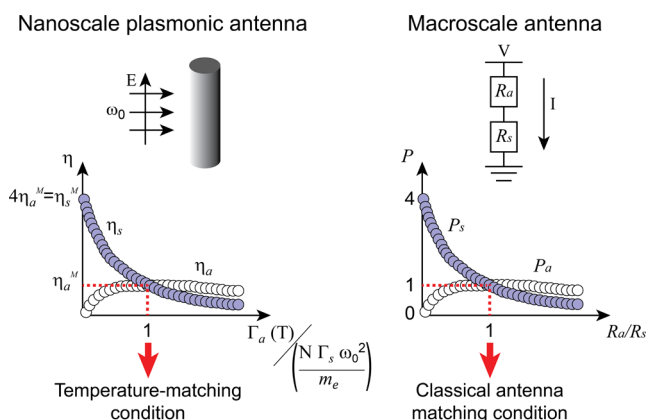


Figure 1. Schematic representation of the similarity between a nanoscale plasmonic antenna (i.e., the nanorods in the present context) and the classical counterpart (for better comparison, the same plots are used for both configurations). In both cases, a condition providing maximum energy absorption can be found, the so-called matching condition. In the nanoscale antenna case, R_s and R_a are the scattering and absorption resistances that account for the reradiated power from the antenna and the absorbed power in the load, respectively. On the other hand, in the plasmonic nanostructure the receiving antenna and the load merge together.

temperature dependence,⁵⁰ it is possible to tune their internal resistance, which at the nanoscale has the same function of the terminal resistance introduced for macroscopic antennas. In this sense, we can realize self-consistent (i.e., without any external load) temperature-matched nanoantennas by selecting under well-defined operational conditions a specific Γ_a^M thus to maximize the absorption efficiency η_a .

Regarding the operational frequencies of these nanoantennas, it is important to underline the relation between their resonant frequency and the temperature maximizing the absorption peak. In order to obtain a decreasing dissipation with increasing

temperature (which is the novel counterintuitive effect at the center of this work), Γ_a^M must be reached at temperatures lower than the operational ones. Given that $\frac{\Gamma_a}{m_e}$ is constant, this means that a proper value for $N\omega_0^2$ should be obtained (see eq 4). One practical possibility is to reduce the resonance frequency of the nanoantennas that, owing to the ω_0 square dependence, can be efficiently achieved by increasing their aspect ratio. This is the approach followed in this work and summarized in Figure 2.

In particular, Figure 2a shows the calculated absorption efficiency η_a for an Au cylinder when the damping coefficient is tuned. The cylinder is as high as $1 \mu\text{m}$ with a 10 nm radius, providing an aspect ratio (AR) equal to 50. The figure shows that while at $\Gamma_a = 0.001$ (gray dot) and $\Gamma_a = 0.1$ (blue dot) the absorption efficiency is far from its maximum, at $\Gamma_a = 0.01$ (green dot) the condition η_a^M is achieved. This result is visualized through dissipation plots, showing maximum heat dissipation at $\Gamma_a = \Gamma_a^M$.

With this in mind, we have performed analytical and numerical calculations for Au nanorods showing different aspect ratios. In particular, Figure 2b–d illustrates the dependence of absorption and scattering efficiencies on the absorption damping coefficient Γ_a for three nanorods placed in air having height equal to $1 \mu\text{m}$ with different radii of 10, 15, and 20 nm. The calculations were performed at the nanorods specific fundamental resonant wavelengths λ_0 , that are 4.35, 3.52, and $3.17 \mu\text{m}$, respectively. These resonant wavelengths were numerically calculated by setting up a predictive model using a commercial FEM software (COMSOL) employing a modified Drude model. A detailed description of the model can be found in refs 50 and 52. In particular, the optical response of Au was described using the permittivity

$$\varepsilon = \varepsilon_0 \left(\varepsilon_{\text{inf}} - \frac{\omega_p^2}{\omega^2 + i\Gamma_a\omega} \right)$$

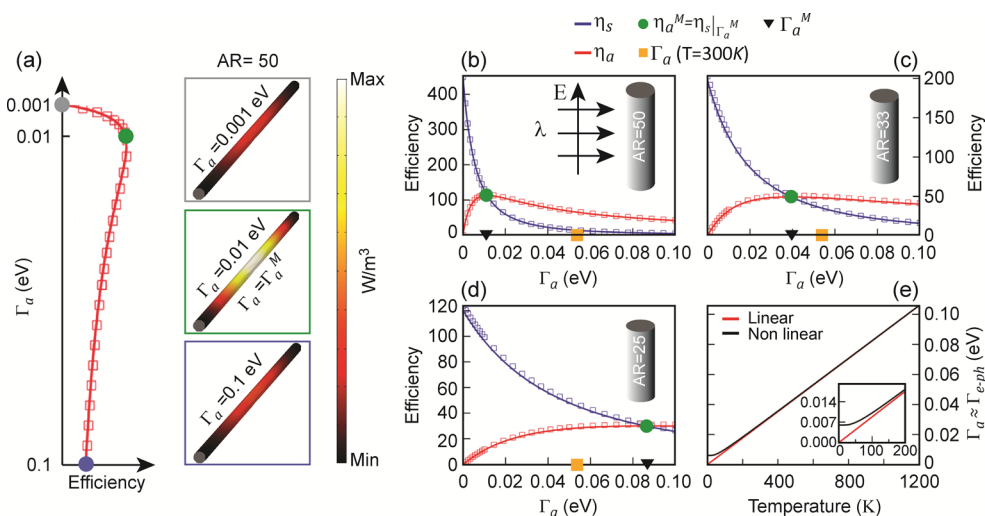


Figure 2. (a) Left: Calculated absorption efficiency as in (b). (a) Right: Three-dimensional colored maps of the nanorods (placed in perspective) describing the heat dissipation at different values of the absorption damping Γ_a . (b–d) Absorption (red) and scattering (blue) efficiencies versus the absorption damping Γ_a for Au cylinders (Drude model) in air (refractive index $n = 1$) with height $H = 1 \mu\text{m}$ and base radius $R = 10, 15, 20$ nm at their resonant wavelength 4.35, 3.52, and $3.17 \mu\text{m}$, respectively. Both analytical calculations (solid lines) and numerical simulations (empty squares) are shown in the figure. The black triangle, the green dot and the yellow square represent the values of Γ_a^M , η_a^M , and Γ_a at 300 K (≈ 0.053 eV), respectively. Insets: Sketches of Au nanorods at different aspect ratio AR (not in scale). For all cases, the light is impinging with the electric field E polarized along the rods axis. (e) Dependence of the electron–phonon damping Γ_{e-ph} on the temperature^{51,64} (black) and in the linear high-temperature limit (red). A closer view at the low temperature range (1–200 K) is shown in the inset.

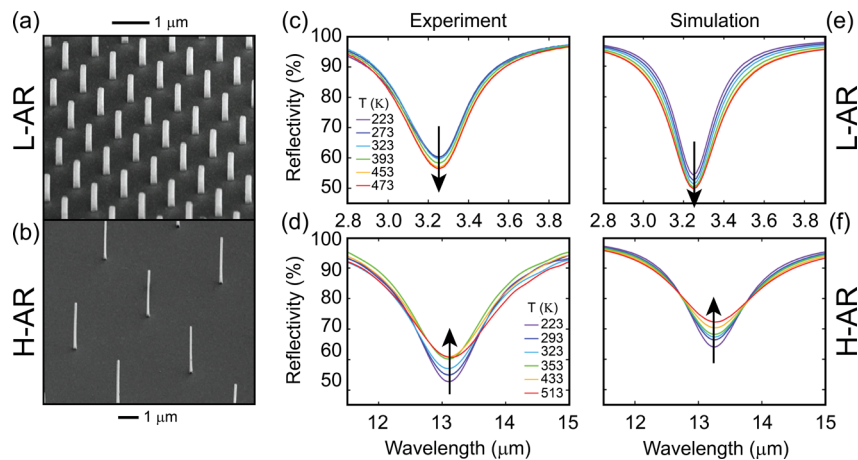


Figure 3. (a) SEM image of the array of L-AR hollow Au antennas used in the experiment. Each antenna is $0.95 \mu\text{m}$ high. The pitch of the unit cell is $1.33 \mu\text{m}$. (b) SEM image of the array of H-AR hollow Au antennas used in the experiment. Each antenna is $3.0 \mu\text{m}$ high. The pitch of the unit cell is $6.0 \mu\text{m}$. For all the antennas, the total radius is $\sim 95 \text{ nm}$ with the hollow part being $\sim 60 \text{ nm}$ and the metallic coating being $\sim 35 \text{ nm}$. The antennas are standing on a substrate covered with an $\sim 50 \text{ nm}$ thick Au layer. Experimental (c,d) and numerical (e,f) reflectivity spectra for the two L-AR (c,e) and H-AR (d,f) antenna arrays at different temperatures from $\sim 200 \text{ K}$ to $\sim 500 \text{ K}$. The arrows signal the reflectivity trend upon temperature increase.

where $\epsilon_{\text{inf}} = 11.5$ takes into account the resonance redshift due to interband transitions⁵⁷ and the absorption damping was considered at room temperature, $\Gamma_a = 53 \text{ meV}$.⁶⁵ Furthermore, a plane wave linearly polarized along the longitudinal nanorods axis is considered. In the figure, the solid lines describe the behavior of the analytical η_a (red) and η_s (blue) as from eq 3. The squared symbols are instead the results from the numerical simulations where Γ_a is tuned between 0 and 0.1 eV. The power associated with the incident radiation was kept small enough to avoid any heating of the nanorods (associated incident electric field equal to 1 V/m). In fact, the additional self-heating due to high-intensities EM fields was investigated in a previous study.^{50,52}

In Figure 2b–d, a remarkable agreement between analytical and numerical results can be observed. In particular, the plots show that the maxima of η_a (green dots) are reached at the crossing point with η_s , that is, at the temperature-matched condition, where the black triangles indicate the values of the absorption damping Γ_a at which the maxima occur. The efficiencies are calculated analytically from eqs 3 and 4 and, as predicted by the aforementioned considerations, the maximum η_a is one-fourth of the maximum η_s (see eq 6). In particular, the number of electrons, N , is calculated as $N = 0.8\rho_{\text{Au}}V$ with $\rho_{\text{Au}} = 5.9 \times 10^{28} \text{ m}^{-3}$ as the Au free electron density and V as the volume of the nanorods. The factor 0.8 was utilized to account for the fact that not all the electrons participate in the plasmonic oscillation due to skin effect.⁵⁶ For comparison, the yellow square defines the value of Γ_a obtained at $T = 300 \text{ K}$ (room temperature). For sake of completeness, it should be noticed that the agreement between analytical and numerical approaches is based on the Drude assumption. Vice versa, the employment of the more complete Drude-Lorentz model into the numerical approach would lead to partial discrepancies with respect to the analytical description such as the impossibility for the absorption to be zero due to the interband transitions. However, the overall behavior would still resemble Figure 2b–d, as shown in the Supporting Figure S1. In the following, where numerical calculations will be compared with experimental results, we shall consider the full form of the Au permittivity.

Another important observation emerging from Figure 2b–d regards the dependence of η_a and η_s on both the aspect ratio and the absorption damping coefficient Γ_a for the three reported geometries. Clearly, we can see that by lowering the nanorods aspect ratio the efficiencies decrease as well while the crossing point between η_a and η_s (i.e., at the matching condition $\eta_a = \eta_s^M = \eta_s^M \Gamma_a^M$) shows a shift toward higher Γ_a . In view of that, the absorption damping coefficient can be used, regardless the aspect ratio, as an extra independent parameter to tune the value of both the efficiencies η_a and η_s . Indeed, as discussed in detail in previous studies,^{50–52} the damping coefficient Γ_a has a relatively sensitive dependence on temperature. In fact, this coefficient takes into account both electron–phonon, $\Gamma_{\text{e-ph}}$, and electron–electron, $\Gamma_{\text{e-e}}$, interactions, that is, $\Gamma_a \approx \Gamma_{\text{e-ph}} + \Gamma_{\text{e-e}}$ where the additional term describing the scattering of electrons with the surface is here neglected as it becomes relevant only in the case of small nanoparticles.⁶⁶ While both e–ph and e–e scattering coefficients depend on the temperature, within the IR-vis range the dominant temperature contribution arises from the electron–phonon interaction because the e–e term exhibits a dominant temperature dependence only at very low frequencies,⁵² hence leading to $\Gamma_a \approx \Gamma_{\text{e-ph}}$. In order to illustrate the effect of temperature on the e–ph scattering coefficient, in Figure 2e the temperature dependence of $\Gamma_{\text{e-ph}}(T)$ is shown from 1 to 1200 K, even though premelting effects have been shown to modify the optical response of metals at relatively low temperatures.^{21,67–70} Experimental outcomes may thus deviate from our predictions upon overcoming temperatures as high as 700 K.⁷¹ The black line derives from the full expression (nonlinear) for the e–ph scattering coefficient obtained by Holstein,⁵³ which assumes a free electrons system with no Umklapp collisions and a single Debye model phonon spectrum. The red line describes instead a linear approximation that can be utilized in the high temperature regime ($T > 200 \text{ K}$). In the Supporting Information we report about both the nonlinear and linear expression for $\Gamma_{\text{e-ph}}(T)$. Regardless of the applied model, Figure 2 provides the following information: the absorption efficiency increases with temperature until the quantity Γ_a reaches the matching value Γ_a^M at which point an opposite behavior is observed in the absorption efficiency response.

This remarkable result suggests the possibility of designing plasmonic devices capable of providing, at the same time, opposite absorption responses with temperature change by simply manufacturing a proper surface patterning. Because absorption directly rules the heat dissipation mechanisms, following this approach it is possible to tailor a system in order to endow it with a given thermal response over a predefined temperature range. This change in temperature can be initiated either by the use of an intense electromagnetic radiation or through the placement of the structure in an extreme low- or high-temperature environment (e.g., a sensor in an oven or a sample in a cryogenic box).

As a proof of concept we have fabricated two different arrays of vertical hollow plasmonic nanoantennas capable of exhibiting opposite dissipative geometrically determined behaviors while their temperature increases from 200 to 500 K. In particular, following the outcome from Figure 2, the samples are designed to have either low (L-AR) or high (H-AR) aspect ratios, as shown in Figure 3, where the SEM images illustrate the plasmonic Au nanoantennas standing on Au surface. L-AR and H-AR nanoantennas are chosen to have a height of 0.95 and 3.0 μm , respectively, and they are fabricated following the procedure presented elsewhere.⁷² The hollowness of these structures, which plays a negligible role at the considered mid-IR wavelengths, depends on the employed fabrication technique. All the nanoantennas have the same total radius (~ 95 nm), which is nominally composed of a hollow part (~ 60 nm) and of a gold coating (~ 35 nm). Furthermore, since the array periodicity is known to influence the optical response^{72–74} of an antenna array, the pitch of the nanoantennas needs to be carefully designed to obtain the desired resonant behavior. The refined interantenna distance is found to be 1.33 and 6.0 μm for the L-AR and H-AR antennas, respectively. Finally, we should stress about the choice of an Au layer overlaying both the antennas and the underneath substrate, together with the condition of constant temperature all over the sample. Indeed, these assumptions allow for the employment of a uniform bulky-like temperature diffusivity which would not have been possible if an effective sample nanopatterning had been considered, especially under laser-induced heating.⁷⁵

We measure the reflectivity spectra of the L-AR and H-AR arrays in resonant conditions by means of a micro-FTIR setup in reflection configuration (details in the Supporting Information, Methods section). Because of the use of a relatively thick gold substrate, the overall transmission of the array is found to be negligible. The experimental results are shown in Figure 3c,d, which well match with the corresponding numerical calculations displayed in Figure 3e,f. In both L-AR and H-AR arrays, the reflectivity changes with temperature. In particular, in the L-AR configuration the experimental reflectivity decreases with the increase of temperature from 223 to 473 K as predicted by the numerical calculations. In fact, as the absorption increases, reflectivity reaches lower values at larger temperatures. This result is coherent with the theoretical calculations reported in Figure 2 where the absorption damping coefficient Γ_a^M (i.e., temperature) calculated for low aspect ratio structures is reached for higher temperatures. Such a behavior is also found in conventional metallic surfaces where the natural roughness can be considered an ensemble of very low aspect ratio structures. The opposite effect is instead observed for the H-AR configuration near resonance, as shown in Figure 3d,f. In fact, by tailoring the height of the antennas and the pitch of the

array, it is possible to attain a patterned structure providing Γ_a^M corresponding to ~ 150 K (that is, $T > T|_{\Gamma_a^M}$) in order to realize an increasing reflectivity with the temperature in the range from 223 to 513 K (see Supporting Figure S2).

These results are highlighted in Figure 4a,b where both the experimental and numerical values of the reflectivity minima at

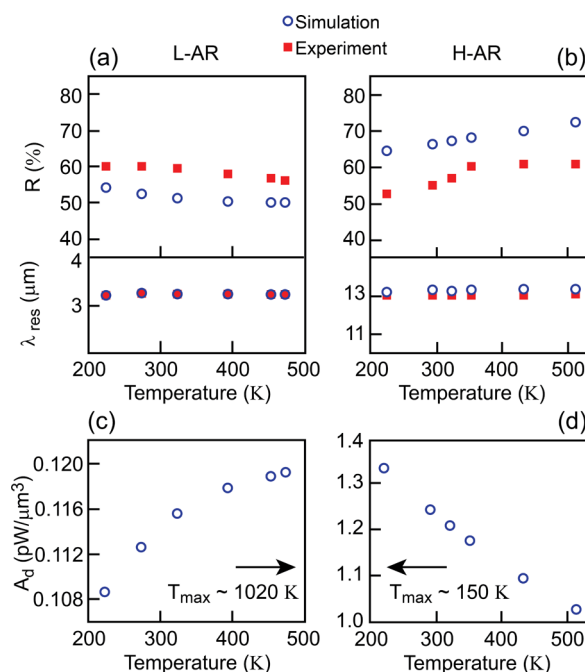


Figure 4. (a,b) Simulations (blue) and experiments (red) for the reflectivity minima dependence on temperature in the case of H-AR (a) and L-AR (b) arrays. The lower parts of the figures show the spectral position of the minima at different temperatures. (c,d) Dependence of the absorption density A_d on temperature for H-AR (c) and L-AR (d) antennas. T_{max} indicates the temperature corresponding to the maximum value of A_d .

different temperatures are shown. Considering that the temperature only weakly influences the real part of the electric permittivity of metals,⁵² the nanoantenna resonances, which mainly depend on it, are not sensitively affected by its change. This situation is shown in the lower inset of Figure 4a,b where the values of the resonant wavelengths associated with the minima in reflection are illustrated. In particular, L-AR and H-AR antenna arrays show constant resonant values equal to ~ 3.25 and ~ 13.2 μm for all the considered temperatures, respectively. These experimental observations on opposite trends in reflectivity confirm our theoretical predictions. Temperature-matched antennas can therefore serve as a platform to realize nanopatterned surfaces with different behavior depending on the temperature regimes.

In order to evaluate the capability of each antenna of the arrays in dissipating heat, we shall introduce the absorption density A_d defined as the total dissipation (Joule heating integrated over the whole antenna) associated with a single antenna and normalized by its volume. Figure 4c,d shows numerical results describing the dependence of A_d on temperature for both the L-AR (c) and the H-AR (d) configurations. Over the temperature range here experimentally investigated, A_d exhibits an increase for the L-AR configuration and a drop for the H-AR case. Once again, this behavior can be understood from eq 4 and Figure 2: higher AR antennas are

characterized by a lower Γ_a^M and thus their absorption density is maximized at lower temperatures. For the H-AR case, in fact, we numerically calculated (see [Supporting Figure S2](#)) the damping coefficient maximizing A_d finding $\Gamma_a^M = 0.015$ eV. This result corresponds to a temperature $T_{\Gamma_a^M} \sim 150$ K, which is below the minimum temperature considered in the experiments. On the other hand, for the L-AR case we found $\Gamma_a^M = 0.089$ eV corresponding to $T_{\Gamma_a^M} \sim 1020$ K, which is well above the maximum temperature employed here. Importantly, it should be noted that in virtue of Kirchhoff's law the emissivity of an extended structure is equal to its absorptivity. This means that our conclusions regarding the absorptivity, (i.e., how it changes with temperature depending on the ratio between radiative and nonradiative losses) can also be applied directly to the emissivity of the structures.⁷⁶ As final and important consideration these findings can be extended to the near-field, whose quantities play a very important role in a number of applications. Indeed, it should be noticed that an increase of temperature lowers the electric field enhancement due to the increase of the imaginary part of the dielectric function (see [Figure S3](#) in the [Supporting Information](#)). This should be taken into account especially for SERS experiments where non-radiative losses can have large effects on Raman scattering.⁷⁷

In conclusion, we have theoretically predicted and experimentally demonstrated how the temperature dependence of the electron–phonon damping term can be utilized to tailor the opto-thermal response of plasmonic nanostructures. We have indeed found that the condition providing maximum absorption efficiency is related to the aspect ratio of the employed nanoantennas. Upon the fulfilment of this condition, we name these structures temperature-matched nanoantennas. In particular, each antenna reaches its maximum light-to-heat conversion efficiency at a specific and fixed temperature, depending on its geometry (number of free electrons) and its resonant frequency. By fabricating two arrays of antennas with different aspect ratio we have experimentally shown that an opposite trend in terms of absorption efficiency vs temperature can be effectively achieved.

It is important to highlight that all these findings can be applied to any resonant system where the absorption damping depends on temperature. In fact, even more exotic behaviors are expected in phase change materials such as Mott insulators, which exhibit a strong temperature dependence of their optical response.⁷⁹ Furthermore, enhanced temperature dependent response can also be expected from structures exhibiting a nonlinear relationship between Γ_a and T .⁸⁰ In this regard, different materials could be employed to aim for alternative opto-thermal response such as shape-memory alloys⁸¹ where their temperature-dependent shape can be exploited for mechano-optical applications at the nanoscale. Another example is provided by undoped or weakly doped graphene nanoribbons that could be investigated in terms of temperature affected optical response by changing the electronic distribution and Fermi level position of the system. In this case, both real and imaginary parts of the dielectric function would be influenced by the temperature, possibly leading to more complex scenarios where both resonance position and damping are tightly linked to thermal processes.

A further interesting application is related to the time rate at which the temperature increases within these systems. In fact, an increasing absorption with temperature allows for a positive feedback in which the larger the increase in the structure temperature, the higher its dissipation becomes, thus leading to

an even higher temperature (such in the case of L-AR antennas shown here). On the other hand, a negative feedback loop can be expected when absorption decreases with temperature (H-AR case). In this case, the temperature increasing process becomes “self-limited” in this sense. These concepts are expected to be useful for optical nanoantennas embedded in microscale electronic circuits, where thermal dissipation evaluation represents a major issue and system cooling becomes a priority.

METHODS

Fabrication of the Nanostructures. Hollow vertical nanostructures were fabricated by slightly modifying the procedure presented in reference [72](#). In order to heat the sample by means of an electric heater, bulk silicon was chosen in place of the silicon nitride suspended membrane used in the cited work, and patterning was performed from the top side. Thin s1813 (Microposit, G2 series) photoresist layers were spin-coated at different speeds to achieve the required thickness, eventually diluting the photopolymer with anisole to obtain 1 μm thick resist. After soft baking samples for 5 min at 110 °C on a hot plate, a thin 50 nm layer of gold was sputtered on the specimen to obtain good conductivity for subsequent ion beam patterning. The core of the fabrication relies on the synergistic action of a focused gallium ion beam, which mills holes in the photoresist with extreme spatial resolution, and the low-energy secondary electrons produced by the interaction between ions and matter, which simultaneously cross-links a thin skin of resist around the patterned hole, throughout the entire depth. After removing gold in its wet etchant and dissolving resist in warm acetone, the insoluble polymeric scaffolds protruding from the silicon were gently rinsed in isopropanol, dried, and coated with metal by sputtering gold on a tilted (45°) rotating sample holder. All samples were then annealed for 2 h at 200 °C to stabilize gold and avoid migration and reassembly of crystalline domains during heating in optical measurements. This latter procedure minimizes the risk of obtaining spurious correlations between effects and the proposed cause together with the benefit of allowing the employment of the bulk Drude-Lorentz⁶⁵ model to numerically describe the illustrated nanostructures. The extension of the electric permittivity to account for temperature change was presented in a previous work.⁵² Finally, differently from other experiments,⁷⁸ the annealing process brought also the advantage of making the nanostructures very resilient to temperature cycling, that is, we could run numerous temperature ramps without any appreciable change in their optical properties as shown in [Figure S4](#).

Optical Characterization. The optical measurements were performed with a commercial micro-FTIR setup (Thermo-Fisher Nicolet iS-50, coupled to its Continuum Infrared microscope), which was used to measure reflectivity from areas of $50 \times 50 \mu\text{m}^2$. The excitation beam was focused (angle of incidence of the beam on the sample is between 15 and 35°) and collected with the same cassegrain condenser objective (15×, numeric aperture 0.55, infinity corrected). Samples were placed in a temperature-controlled stage (Linkam Scientific THMS600, temperature specifications ranging from liquid nitrogen to +600 °C) and kept in controlled atmosphere by an overpressure of nitrogen gas in the sealed control chamber. Illumination was allowed through a ZnSe window (transparent to IR radiation up to 20 μm wavelengths). Using these precautions to avoid any time-dependent fluctuations in

absorption and reflection intensity, for instance, connected to heated air turbulence, or ice crystal formation from atmospheric humidity condensation, we managed to acquire spectra in the ranges between $-80\text{ }^{\circ}\text{C}$ and $+200\text{ }^{\circ}\text{C}$. At lower temperatures, microcrystals appeared despite the protected nitrogen atmosphere, whereas for temperatures higher than $240\text{ }^{\circ}\text{C}$ the antennas inner polymeric scaffold starts degrading, resulting in the structure bending and thus it showing dramatic changes in the spectra. All experiments were carried on cycling twice from cold to hot with a waiting time of 10 min to ensure the thermalisation of the system at each temperature in order to verify that no changes in spectra were observed. This rules out the possibility that differences in reflectivity magnitude may be connected to irreversible plastic deformations of the structures due to heating. At last, all measurements were normalized to a reference spectrum collected on a flat gold area coplanar to the patterned structures. Every measurement is the average of 128 acquired spectra with a resolution of $4\text{ points}/\text{cm}^{-1}$.

Numerical Calculations. The electromagnetic calculations were performed using the commercial software COMSOL Multiphysics. Calculations for the single $1\text{ }\mu\text{m}$ long nanorods case were carried out by placing the single antenna (described by a Drude-like electric permittivity) in a background medium (refractive index 1) represented by a $3\text{ }\mu\text{m}$ diameter sphere. The simulation domain was truncated by employing 350 nm thick perfectly matched layers (PML). The incident radiation was represented by a plane wave linearly polarized along the nanorods longitudinal axes and directed perpendicularly to them. The resonant frequency was first calculated for each nanorods' base radius by detecting the absorption peak using 53 meV as absorption damping (corresponding to the room temperature case, 300 K). Absorption and scattering plots were calculated at the correspondent resonance frequencies for each antenna by tuning the damping coefficient between 0 and 0.1 eV . Only one-fourth of the total geometry was included in the simulations given the 4-fold symmetry of the system. The fabricated and experimentally characterized samples were modeled as infinite three-dimensional arrays of antennas where Floquet periodic conditions were set in the planar directions to define the rectangular unit cells. Input radiation was accounted for by setting electromagnetic ports at the top surface of the modeled domain, on the interior side of the PML layer, setting an angle of incidence of 20° which is within the experimental range. Analytical calculations were performed using Mathematica.

■ ASSOCIATED CONTENT

● Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: [10.1021/acs.nanolett.7b02131](https://doi.org/10.1021/acs.nanolett.7b02131).

Comparison between classical and plasmonic matching; derivative of absorption and scattering efficiencies with respect to absorption damping; Electron–phonon scattering term dependence on temperature; absorption and scattering efficiency versus damping using the complete Au dielectric function; dependence of absorption density on damping: H-AR and L-AR cases; supporting references (PDF)

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Notes

The authors declare no competing financial interest.

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